Name:

Address:

Energy Assessment Report

Add house illustration with solar panels

Project location	**Lagos, Nigeria**
Property type	Hotel
Assessment date	8 March 2026
Primary objective	Reduce diesel use and improve reliability

Recommended PV	Battery	Annual savings	Payback
28 kWp	**40 kWh**	**NGN 7,800,000**	**4.2 years**
Preliminary sizing	Backup support	Indicative	Simple payback

SolarVy

Name:

Address:

Executive summary

The Solarvy platform evaluated the site as a **hotel** in **Lagos, Nigeria** with a primary objective to **reduce diesel use and improve reliability**. Based on the sample inputs, the site appears well suited to a **hybrid solar plus battery solution** because generator dependence is meaningful, outage exposure is material, and the estimated operating profile suggests strong value in shifting daytime demand away from diesel.

The model indicates an example configuration of **28 kWp solar PV**, **40 kWh battery storage**, and a **25 kW hybrid inverter**. At this level, the strongest financial value driver is the reduction of diesel-backed electricity, with indicative annual savings of **NGN 7,800,000** and a simple payback of approximately **4.2 years**.

Key finding	Example result
Estimated annual demand	216,000 kWh/year
Recommended PV	28 kWp
Recommended battery	40 kWh
Estimated diesel reduction	12,400 litres/year
Estimated annual savings	NGN 7,800,000
Estimated simple payback	4.2 years

Important note: this report is preliminary. Final design, equipment selection, and performance should be validated through a more detailed engineering review before procurement or installation.

SolarVy

Name:

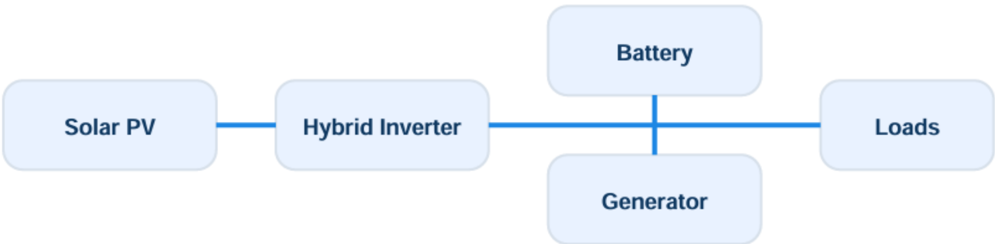
Address:

Energy profile

Metric	Example value
Average monthly energy use	18,000 kWh/month
Estimated annual demand	216,000 kWh/year
Estimated peak demand	45 kW
Typical outage exposure	6 hours/day
Primary energy objective	Reduce diesel use and improve reliability

The example load profile suggests a site with meaningful daily demand and recurring outage exposure. This kind of operating environment typically supports a hybrid system case because solar generation can offset daytime energy use, while battery storage can reduce generator runtime and support critical loads during interruptions.

Illustrative system architecture



Name:

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Recommended system and financial analysis

Component	Recommendation
Solar PV capacity	28 kWp
Battery storage	40 kWh
Hybrid inverter	25 kW
Annual PV generation	40,600 kWh/year
Usable solar energy	32,500 kWh/year

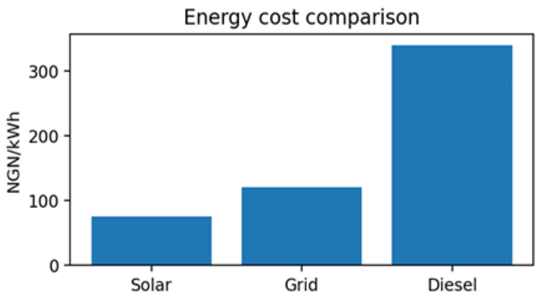
Metric	Example result
Total estimated system cost	NGN 32,000,000
Gross annual savings	NGN 8,100,000
Annual O&M allowance	NGN 300,000
Net annual savings	NGN 7,800,000
Simple payback	4.2 years
NPV	NGN 18,500,000
IRR	21%

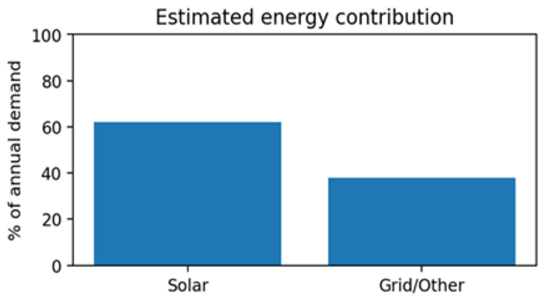
Name:

Address:

Hybrid performance and value drivers

Performance metric	Example result
Diesel saved	12,400 litres/year
Diesel displacement value	NGN 14,880,000
Energy independence ratio	62%
Estimated CO_2 reduction	33 tonnes/year





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Recommendation narrative and next steps

Recommendation insight

A hybrid solar plus battery system is recommended because generator dependence appears significant and the example model indicates strong fuel-saving potential. Battery storage is included primarily to reduce generator runtime and support continuity of supply during outages. The recommended system should be treated as a preliminary assessment rather than a final design package.

Recommended next step	Why it matters
Request detailed technical review	Validate load assumptions, battery sizing, and real operating profile before procurement.
Obtain installer quotations	Compare implementation options against the preliminary system recommendation.
Review commercial assumptions	Check tariff, diesel price, and operating profile against actual site data.

CTA

Request a detailed review from SAEK Energy or Get matched with supplier and installer option

Disclaimer: This report is an indicative assessment generated from example inputs and simplified assumptions. Final design, procurement, performance, and financial outcomes may vary.